# Hongchao Zhang

Google Scholar | Personal Website

Email: hongchao@wustl.edu
Mobile: (508) 797-2793


Artificial intelligence (AI) now transforms not only software but also the physical world. My research asks how we can trust AI-enabled systems when their decisions have physical consequences. I develop formal verification and safe control methods with provable guarantees for robots and critical infrastructure, including the electric and water networks that AI data centers rely on.

## EDUCATION

- **Washington University in St. Louis** — St. Louis, MO
  *Ph.D., Electrical Engineering (Advisor: Prof. Andrew Clark)* — *2020 – 2025*
  - Thesis: "Resilient Safe Control of Autonomous Systems"
- **Worcester Polytechnic Institute** — Worcester, MA
  *M.S., Electrical & Computer Engineering* — *2018 – 2020*
- **Nanjing University of Aeronautics and Astronautics** — Nanjing, China
  *B.E., Automation Engineering* — *2014 – 2018*

## PROFESSIONAL EXPERIENCE

- **Vanderbilt University** — Nashville, TN
  *Postdoctoral Scholar, Institute for Convergent Software Integrated Systems (ICONS)* — *Sep 2025 – Present*
  - Research at the intersection of AI, cyber-physical systems, and formal methods with Prof. Taylor T. Johnson, Prof. Meiyi Ma, and Prof. Ahmad F. Taha.
- **Washington University in St. Louis** — St. Louis, MO
  *Graduate Research Assistant* — *Aug 2022 – Aug 2025*
- **Worcester Polytechnic Institute** — Worcester, MA
  *Graduate Research Assistant* — *Aug 2019 – Aug 2022*

## JOURNAL PAPER

[J1] **[IEEE TAC] Zhang, Hongchao**, Zhouchi Li, and Andrew Clark. *"Safe Control for Nonlinear Systems under Faults and Attacks via Control Barrier Functions." IEEE Transactions on Automatic Control*, vol. 70, no. 11, pp. 7334-7349, Nov. 2025, doi: 10.1109/TAC.2025.3571338.

[J2] **[IEEE TAC] Zhang, Hongchao**, Manan Tayal, Jackson Cox, Pushpak Jagtap, Shishir Kolathaya, and Andrew Clark. *"Stochastic Neural Control Barrier Functions." IEEE Transactions on Automatic Control*, to appear March 2027.

### Under Review

[J-U1] **Zhang, Hongchao**, Mohamad H. Kazma, Meiyi Ma, Taylor T. Johnson, and Ahmad F. Taha. *"Verification and Forward Invariance of Control Barrier Functions for Differential-Algebraic Systems." International Journal of Robust and Nonlinear Control*, under review. arXiv:2603.13509 (2026).

## CONFERENCE PAPER

[C1] **[IJCAI]** Chen, Zirong, **Hongchao Zhang**, and Meiyi Ma. *"PACE: A Personalized Adaptive Curriculum Engine for 9-1-1 Call-taker Training."* In *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence*, pp. 7093–7101, 2026. doi: 10.24963/ijcai.2026/789.

[C2] **[ICML]** Liu, Yilin, **Hongchao Zhang**, Ahmad Taha, Taylor T Johnson, and Meiyi Ma. *"Modeling Spectral Energy Shifts in Spatio-Temporal Graph Anomaly Detection."* In *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306, pp. 79840–79854, 2026.

[C3] **[IEEE CDC] Zhang, Hongchao**, Mohamad H. Kazma, Ben Wooding, Meiyi Ma, Taylor T. Johnson, and Ahmad F. Taha. *"Control Barrier Function Synthesis for Differential-Algebraic Systems."* In *IEEE Conference on Decision and Control*, 2026, to appear.

[C4] **[IEEE CDC]** Kazma, Mohamad H., **Hongchao Zhang**, Abdallah A. Albustami, Ben Wooding, Meiyi Ma, Taylor T. Johnson, and Ahmad F. Taha. *"Safe Control of Semi-Explicit Differential-Algebraic Systems: Control Barrier Functions with SOS Verification."* In *IEEE Conference on Decision and Control*, 2026, to appear.

[C5] **[IEEE CDC]** Wooding, Ben, Mohamad H. Kazma, **Hongchao Zhang**, Ahmad F. Taha, Meiyi Ma, and Taylor T. Johnson. *"Safe Autonomous Takeover for Unknown Descriptor Systems."* In *IEEE Conference on Decision and Control*, 2026, to appear.

[C6] **[SAIV]** Wooding, Ben, Kiersten Brennan, Anne M. Tumlin, **Hongchao Zhang**, and Taylor T. Johnson. *"A Self-correcting Neuro-symbolic AI Reasoning Framework."* In *AI Verification (SAIV 2026)*, LNCS 16831, pp. 191–211. Springer, 2027 (published online 2026). doi: 10.1007/978-3-032-32357-6_9.

[C7] **[NeuS]** Wooding, Ben, **Hongchao Zhang**, Taylor T. Johnson, and Abolfazl Lavaei. *"k-Inductive Neural Barrier Certificates for Unknown Nonlinear Dynamics."* In *3rd International Conference on Neuro-symbolic Systems (NeuS'26)*, PMLR, 2026.

[C8] **[ATVA]** Tumlin, Anne M.*, Samuel Sasaki*, Ben Wooding, Diego Manzanas Lopez, Muhammad Usama Zubair, Navid Hashemi, **Hongchao Zhang**, Waseem Abbas, Ipek Oguz, Meiyi Ma, and Taylor T. Johnson. *"NNV3: Expanding Neural Network Verification to New Architectures and Domains."* In *24th International Symposium on Automated Technology for Verification and Analysis (ATVA)*, Springer, 2026, to appear. (*Equal contribution.)

[C9] **[FMAS]** Serbinowska, Serena S., Ben Wooding, Diego Manzanas Lopez, Luke Nam, Anne M. Tumlin, Nicholas K. Potteiger, **Hongchao Zhang**, and Taylor T. Johnson. *"BehaVerify: A Formal Verification Tool for Behavior Trees."* In *8th International Workshop on Formal Methods for Autonomous Systems (FMAS)*, 2026, to appear.

[C10] **[ACM CCS]** Wang, Jinwen*, **Hongchao Zhang***, Chuanrui Jiang, Andrew Clark, and Ning Zhang. *"ConTest: Taming the Cyber-physical Input Space in Fuzz Testing with Control Theory."* In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2025. (*Equal contribution.)

[C11] **[NeurIPS] Zhang, Hongchao**, Zhizhen Qin, Sicun Gao, and Andrew Clark. *"SEEV: Synthesis with Efficient Exact Verification for ReLU Neural Barrier Functions."* In *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[C12] **[IEEE CDC]** Dai, Hongkai, Chuanrui Jiang, **Hongchao Zhang**, and Andrew Clark. *"Verification and Synthesis of Compatible Control Lyapunov and Control Barrier Functions."* In *IEEE Conference on Decision and Control*, 8178–8185, 2024.

[C13] **[IEEE CDC]** Tayal, Manan, **Hongchao Zhang**, Pushpak Jagtap, Andrew Clark, and Shishir Kolathaya. *"Learning a Formally Verified Control Barrier Function in Stochastic Environment."* In *IEEE Conference on Decision and Control*, 4098–4104, 2024.

[C14] **[IEEE ICAA]** Niu, Luyao, **Hongchao Zhang**, Dinuka Sahabandu, Bhaskar Ramasubramanian, Andrew Clark, and Radha Poovendran. *"Who is Responsible? Explaining Safety Violations in Multi-Agent Cyber-Physical Systems."* In *IEEE International Conference on Assured Autonomy*, 2024.

[C15] **[IEEE ICRA] Zhang, Hongchao**, Luyao Niu, Andrew Clark, and Radha Poovendran. *"Fault Tolerant Neural Control Barrier Functions for Robotic Systems under Sensor Faults and Attacks."* In *IEEE International Conference on Robotics and Automation*, 9901–9907, 2024.

[C16] **[NeurIPS] Zhang, Hongchao**, Junlin Wu, Yevgeniy Vorobeychik, and Andrew Clark. *"Exact Verification of ReLU Neural Control Barrier Functions."* In *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[C17] **[IEEE CDC] Zhang, Hongchao**, Zhouchi Li, Hongkai Dai, and Andrew Clark. *"Efficient Sum of Squares-Based Verification and Construction of Control Barrier Functions by Sampling on Algebraic Varieties."* In *IEEE Conference on Decision and Control*, 5384–5391, 2023.

[C18] **[VehicleSec] Zhang, Hongchao**, Zhouchi Li, Shiyu Cheng, and Andrew Clark. *"Cooperative Perception for Safe Control of Autonomous Vehicles under LiDAR Spoofing Attacks."* In *ISOC Symposium on Vehicle Security and Privacy*, 2023. **GM AutoDriving Security Award.**

[C19] **[IEEE CDC] Zhang, Hongchao**, Shiyu Cheng, Luyao Niu, and Andrew Clark. *"Barrier Certificate based Safe Control for LiDAR-based Systems under Sensor Faults and Attacks."* In *IEEE Conference on Decision and Control*, 2256–2263, 2022.

[C20] **[IEEE CDC]** Niu, Luyao, **Hongchao Zhang**, and Andrew Clark. *"Safety-Critical Control Synthesis for Unknown Sampled-Data Systems via Control Barrier Functions."* In *IEEE Conference on Decision and Control*, 6806–6813, 2021.

[C21] **[IEEE ICRA] Zhang, Hongchao**, Zhouchi Li, and Andrew Clark. *"Model-based Reinforcement Learning with Provable Safety Guarantees via Control Barrier Functions."* In *IEEE International Conference on Robotics and Automation*, 792–798, 2021.

[C22] **[IEEE CDC]** Clark, Andrew, Zhouchi Li, and **Hongchao Zhang**. *"Control Barrier Functions for Safe CPS Under Sensor Faults and Attacks."* In *IEEE Conference on Decision and Control*, 796–803, 2020.

### Under Review

[C-U1] Zhang, Yunuo, Nathaniel S Keplinger, **Hongchao Zhang**, Baiting Luo, Ayan Mukhopadhyay, and Abhishek Dubey. *"Semi-Implicit Belief Approximation with Learned Transport and Reweighting for POMDPs." ICLR 2027*, under review.

[C-U2] **Zhang, Hongchao**, Guocheng He, Ben Wooding, Ahmad Taha, Taylor T Johnson, and Meiyi Ma. *"Error Localization and Steering of Language Model Quantitative Reasoning." ICLR 2027*, under review.

[C-U3] Liu, Yilin, **Hongchao Zhang**, Ahmad Taha, Taylor T Johnson, and Meiyi Ma. *"Generalist Graph Anomaly Detection via a Spectral Vocabulary." ICLR 2027*, under review.

[C-U4] Wooding, Ben, Anne Marie Tumlin, Samuel Sasaki, **Hongchao Zhang**, and Taylor T Johnson. *"Input-Agnostic Probabilistic Neural Network Verification for Vision, Speech and Depth." AAAI 2027*, under review.

[C-U5] He, Guocheng, Lingwen Deng, Ahmad Taha, Taylor T Johnson, **Hongchao Zhang**, and Meiyi Ma. *"Formal Method Guided Multi-Agent Optimization Formulation." AAMAS 2027*, under review.

[C-U6] Wooding, Ben*, **Hongchao Zhang***, and Taylor T. Johnson. *"Safe Flight in Cluttered Indoor Environments: A Verified Union of Neural Control Barrier Functions." IEEE International Conference on Robotics and Automation (ICRA)*, under review. (*Equal contribution.)

[C-U7] Atiq, Aarij, Samuel Sasaki, Ben Wooding, Meiyi Ma, Ahmad Taha, Taylor Johnson, and **Hongchao Zhang**. *"Robustness Verification of First-to-Fire Spiking Neural Networks." TACAS 2027*, under review.

## Funding and Awards

- **Vanderbilt Center for Sustainability, Energy, and Climate Seed Project Fund Award** ($200,000 total) 2026
  *Reliable AI and Critical Infrastructure: Compliance, Resilience, and Explainability in Electric and Water Networks.*
  Taylor T. Johnson, Meiyi Ma, Ahmad F. Taha, Mark Williams, and **Hongchao Zhang**.
- **National Science Foundation (NSF), CMMI-2418806** 2024
  Provided neural-symbolic verification supporting the grant.
- General Motors AutoDriving Security Award 2023
  ISOC Symposium on Vehicle Security (VehicleSec @ NDSS)
- Shao-Fang and Tsu-Chin Lee Endowed Fellowship 2022
  Washington University in St. Louis

## Teaching

- **CS 3892/5892, Formal Methods: Safe and Trustworthy AI** Fall 2026
  *Vanderbilt University. Guest Lecturer. Specifications and runtime enforcement.*
- **CS 6378, Cyber-Physical Artificial Intelligence** Fall 2026
  *Vanderbilt University. Guest Lecturer. Human–agent collaboration, from specifications to acceptance.*
- **CS 6315, Automated Verification** Spring 2026
  *Vanderbilt University. Guest Lecturer.*
- **CS 3892/5892, Projects in Cloud Computing** Fall 2025
  *Vanderbilt University. Guest Lecturer.*
- **ESE 559, Learning and Planning in Robotics** Spring 2025
  *Washington University in St. Louis. Assistant Instructor.*
- **ESE 2180, Linear Algebra and Component Analysis** Fall 2024
  *Washington University in St. Louis. Leading Assistant Instructor.*

## Invited Talks

- **Vanderbilt University** Sep 2025
  Host: Thomas Beckers. *"Resilient Safe Control of Autonomous Systems."*
- **University of Michigan** Feb 2025
  Host: Necmiye Ozay. *"Resilient Safe Control of Autonomous Systems."*
- **Washington University in St. Louis** Apr 2024
  Host: Center for Trustworthy AI in CPS. *"Cooperative Perception for Safe Control of Autonomous Vehicles under LiDAR Spoofing Attacks."*

## Mentoring

- Vanderbilt University Graduate Directed Research: **Yilin Liu** (2025); coauthor on [C2, C-U3], **Guocheng He** coauthor on [C-U2, C-U5].
- Vanderbilt University Undergraduate Directed Research: **Aarij Atiq** (2025); coauthor on [C-U7].
- Washington University Research Experience for Undergraduates (REU): **Ziyi Peng** (2025); **Mario Rodriguez-Montoya**, **Dzenan Zecevic** (2024); **Nicolas Hernandez**, **Keyron Linarez** (2023)
- Washington University Graduate Directed Research: **Mark Charnot**, **Seifeldin I. Elkhashab** (2023)
- Worcester Polytechnic Institute Major Qualifying Project (MQP): **Noelle Johnson** (2022); **Minh Le** and **Christopher Letherbarrow** (2019)

## Service

- **HSCC+ICCPS 2027** Artifact / Repeatability Evaluation Co-Chair.
- **Conference reviewing:** AAAI (2026, 2027); NeurIPS (2025, 2026); ICLR (2026); IEEE CDC (2022–2026); ACC (2026); IROS (2026); EMSOFT (2026); CHASE (2026); IEEE RTSS (2026); QEST/FORMATS (2026).
- **Journal reviewing:** IEEE TCNS (2025, 2026); IEEE TAC (2023–2026); IEEE T-IFS (2023–2025); ACM TCPS (2023–2025); IEEE TRO (2025); IEEE Robotics and Automation Letters (2025, 2026); IEEE Transactions on Intelligent Vehicles (2026).